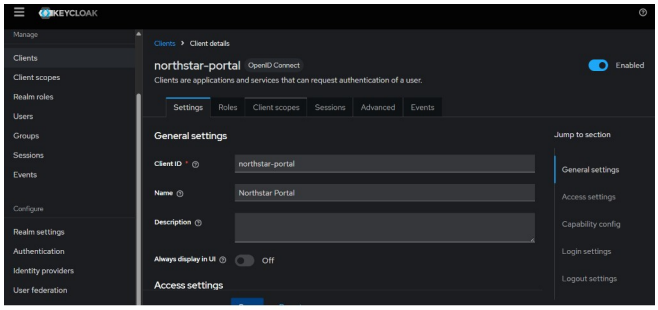
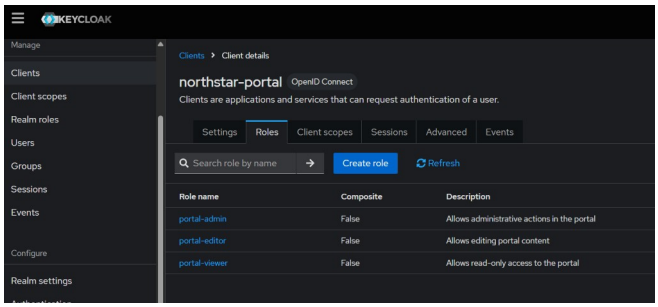
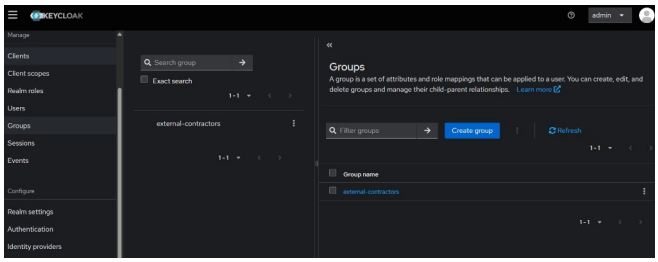
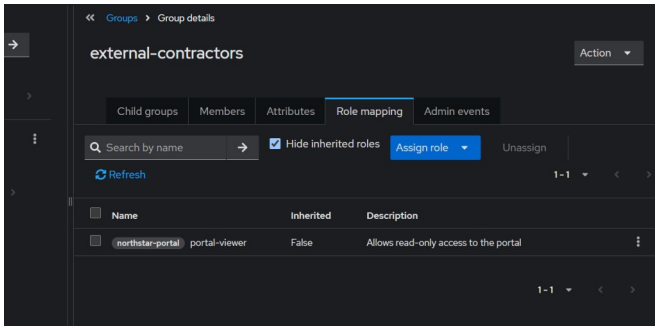
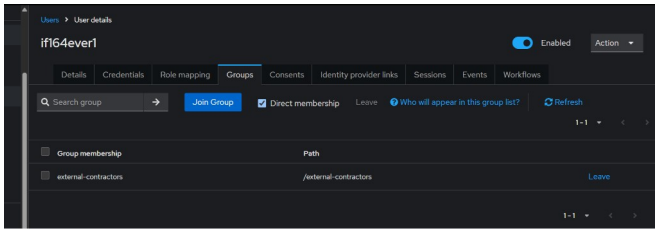
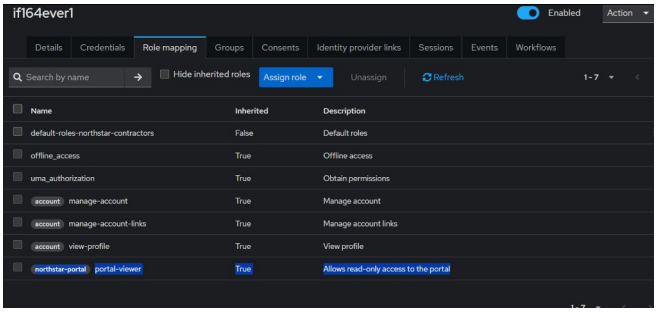
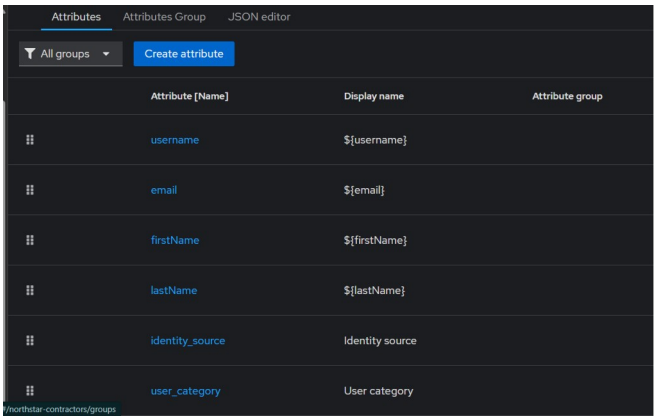
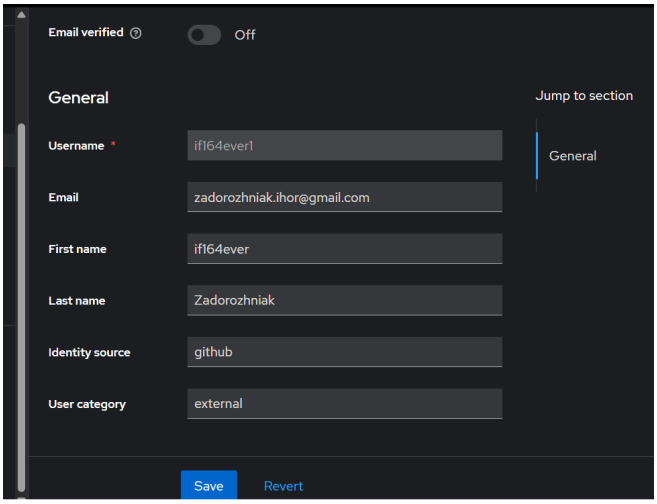
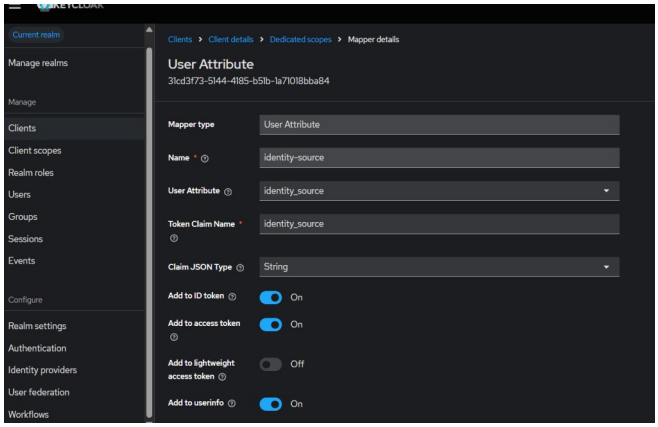
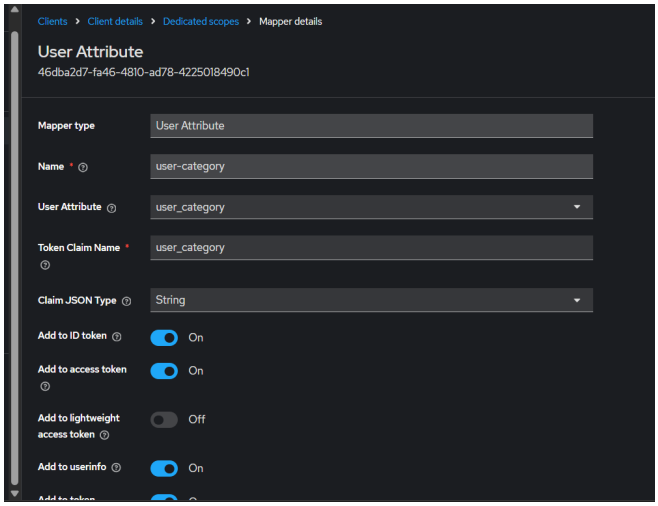
The `northstar-portal` OpenID Connect client was created in the `northstar-contractors` realm with Standard Flow enabled. This client represents the internal portal.	
Screenshot 2 — Client Roles Three client roles were created: `portal-viewer`, `portal-editor`, and `portal-admin`. These roles define read-only, editing, and administrative access to the portal.	
Screenshot 3 — External Contractors Group The `external-contractors` group was created to organize external users. This group allows access permissions to be managed through group membership.	
Screenshot 4 — Group Role Mapping The `portal-viewer` role from the `northstar-portal` client was assigned to the `external-contractors` group. Members of this group inherit the portal's read-only role.	
Screenshot 5 — User Group Membership The GitHub-authenticated user was added to the `external-contractors` group. This membership grants the user the roles assigned to the group.	
Screenshot 6 — Inherited User Role	

The user’s effective role mappings were checked to confirm that `portal-viewer` was inherited through group membership. The role was not assigned directly to the user.	
Screenshot 7 — User Profile Attributes The `identity_source` and `user_category` attributes were created in the realm’s User Profile settings. Both attributes were configured as optional fields that administrators can view and edit.	
Screenshot 8 — User Attribute Values The user’s `identity_source` attribute was set to `github`, and `user_category` was set to `external`. These values were saved locally in Keycloak to describe the user’s identity source and category.	

Screenshot 9 — Identity Source Protocol Mapper A User Attribute protocol mapper was configured to include `identity_source` as a string claim. The mapper adds this value to the access token, ID token, and UserInfo response.	
Screenshot 10 — User Category Protocol Mapper A second User Attribute protocol mapper was configured to include `user_category` as a string claim. The mapper adds this value to the access token, ID token, and UserInfo response.	
Screenshot 11 — Token Evaluation The client's Evaluate feature was used to generate an example access token for the GitHub user. The payload contained `identity_source=github`, `user_category=external`, and the inherited `portal-viewer` role under `resource_access.northstar-portal.roles`.	